

Are cells the company they keep?

Neighborhood & Expression Informed Latent-Variable Estimation with Deep Kernel Learning

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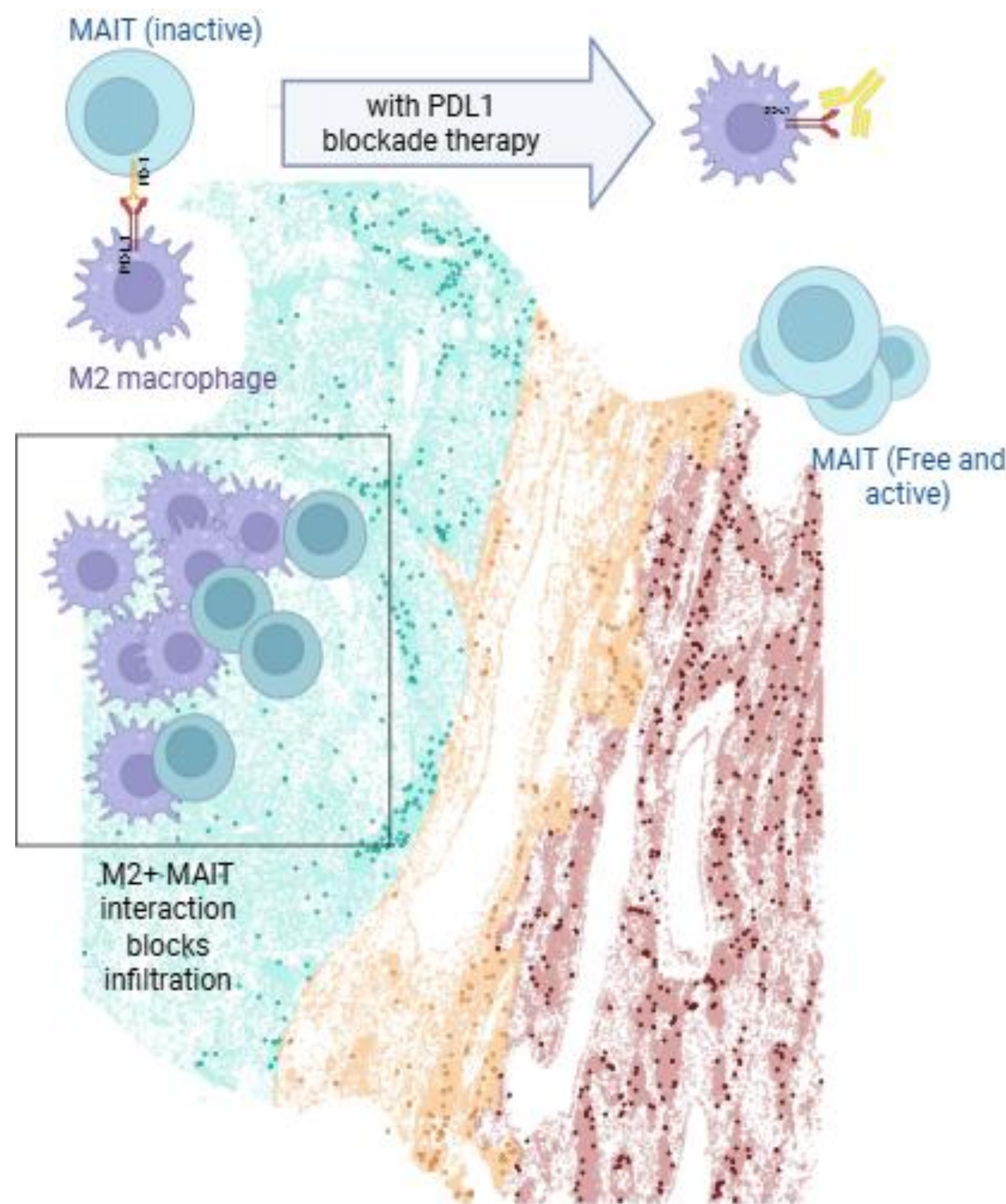
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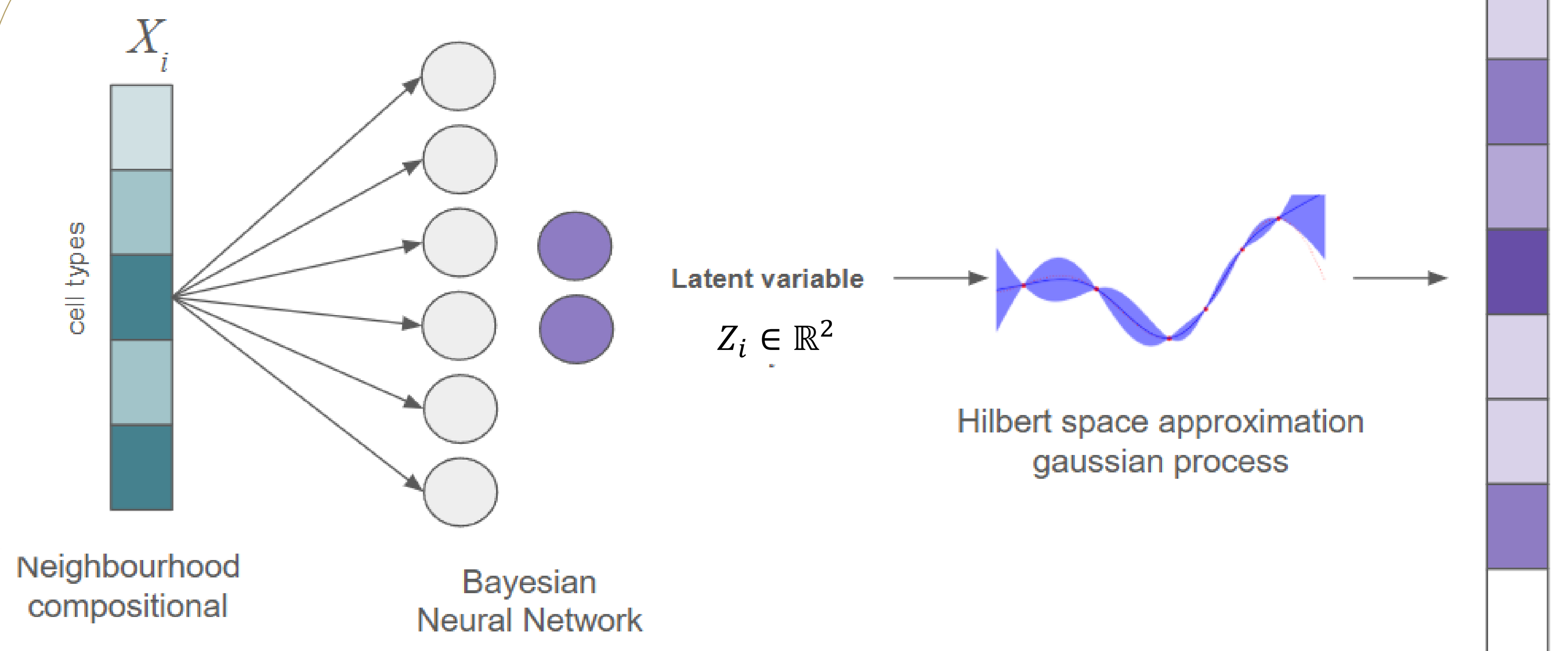
How does the **cellular neighborhood** influence **MAIT cell behavior** across different **tissue zones** in hepatocellular carcinoma?

Biological Context & Data



- High-multiplex, single-cell, spatial proteomics using CODEX imaging resulting in 34 markers [1]
- Human patient samples ($n = 20$) of hepatocellular carcinoma (HCC)
- 3 distinct tissue zones: tumor core, rim and normal liver

Model Outline



Bayesian Neural Network (BNN):

- Weights and biases are not deterministic, but distributions
- Input: **k-nearest neighborhood** with **exponential distance decay**
- To ensure signal flow and avoid shrinkage, we test multiple prior distributions and configuration of weights

Hilbert-space Gaussian process (HSGP)

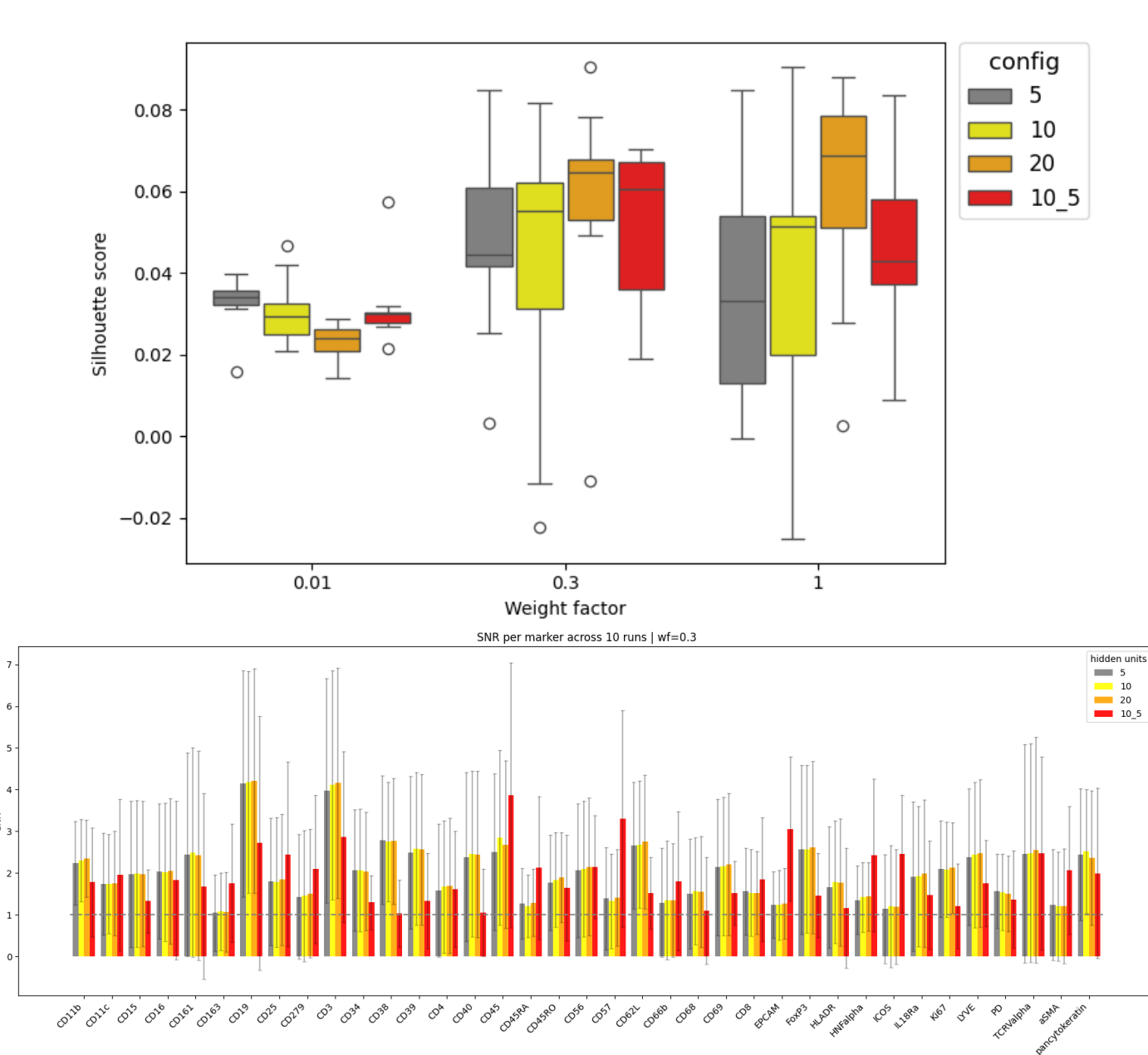
- Key assumption: gene expression varies smoothly over latent embedding
- GP's scale poorly with data size $O(N^3)$
- Gene-specific signal α_g^2 and noise σ_g^2

Evaluation metrics

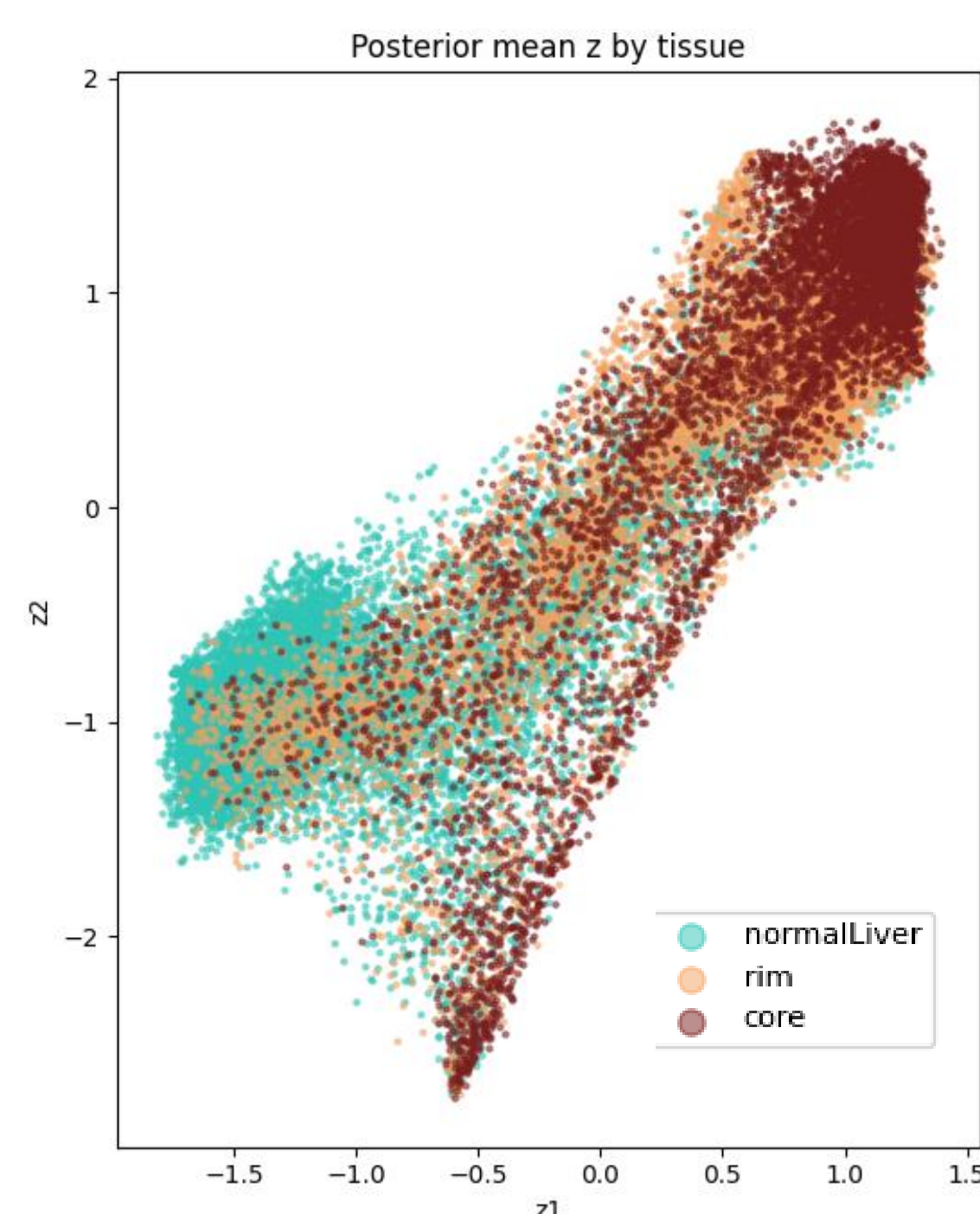
- Silhouette score for tissue separability
- "Signal to noise ratio" (α_g/σ_g) for gene expression modelling
- Variance preserved across encoder

Results

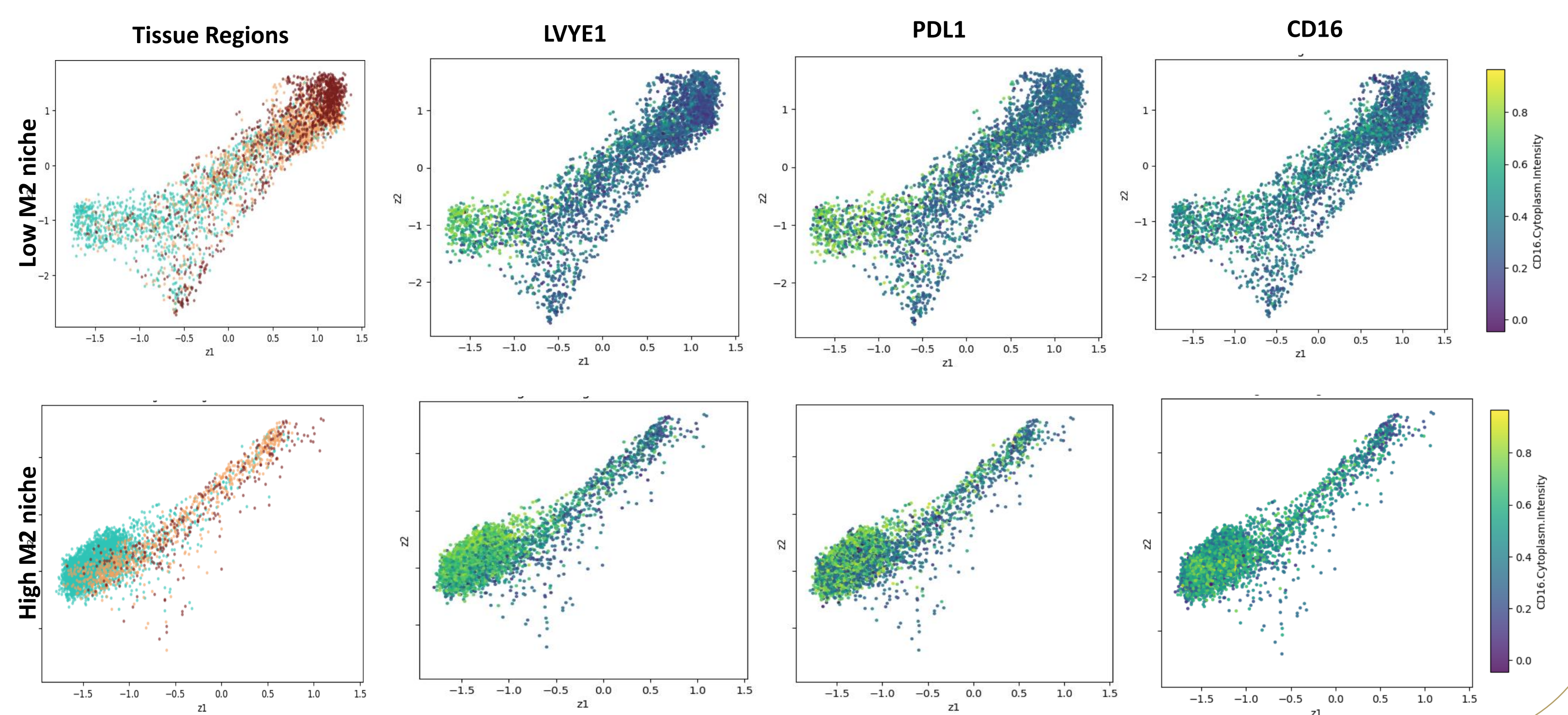
1. Architecture Ablations



2. Inferred latent Space (z)



3. Stratification of z by M2 macrophage density



Conclusion

- Non-linear networks yield latent spaces that are **better separated** in **tissue regions** than those generated by linear models.
- Systematic testing of network width, and weight initializations reveals that **wider** BNN architectures yield **higher separability** and **lower SNR**
- The latent space groups cells by tissue region, revealing that some markers are differentially expressed when transitioning from low-density to high-density M2 macrophage neighborhoods.

Outlook

- Potential improvement with radial neighborhood definition
- Use single-cell spatial transcriptomics for broader understanding of cells
- Explore more architectures for non-linear embedder: deeper neural networks, graph neural networks,
- Explicitly model batch and patient effects in the parameterization of the HSGP
- Explicitly model gene-gene correlations

References

[1] Ruf et al., Cell, 2023. Tumor-associated macrophages trigger MAIT cell dysfunction at the HCC invasive margin.



LinkedIn



Code